Practical implementation

1. Stage 1

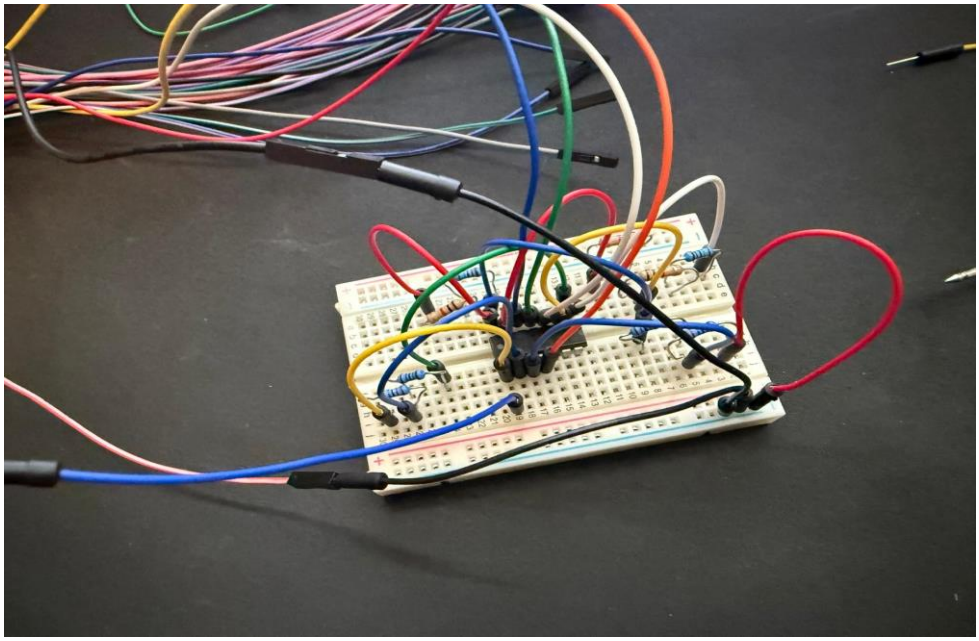

Figure 44. Stage 1 on breadboard implementation (side view)

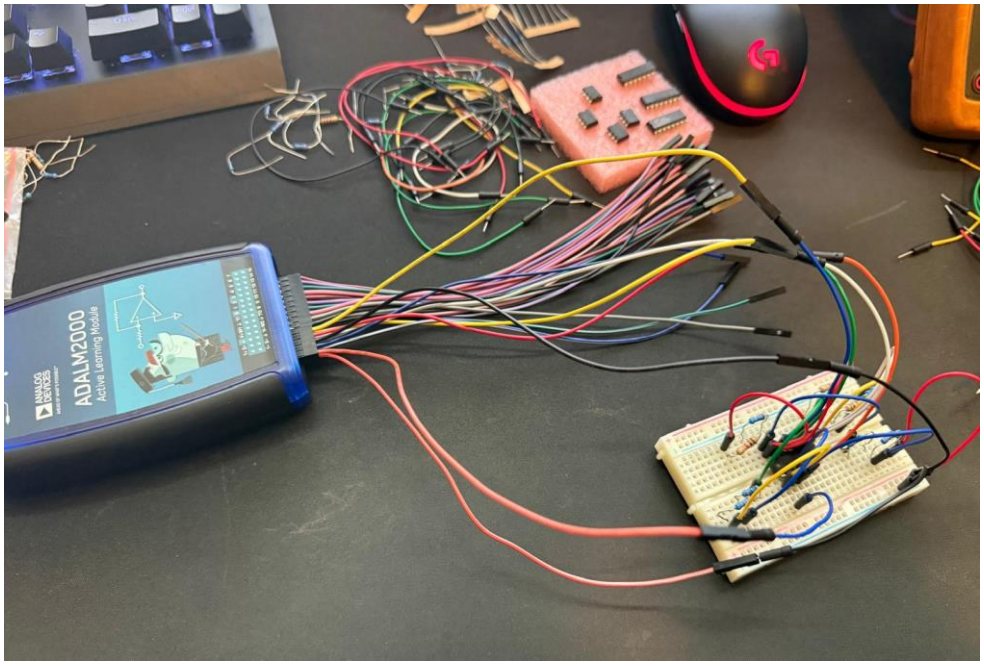

Figure 45. Stage 1 on breadboard implementation (distant view)

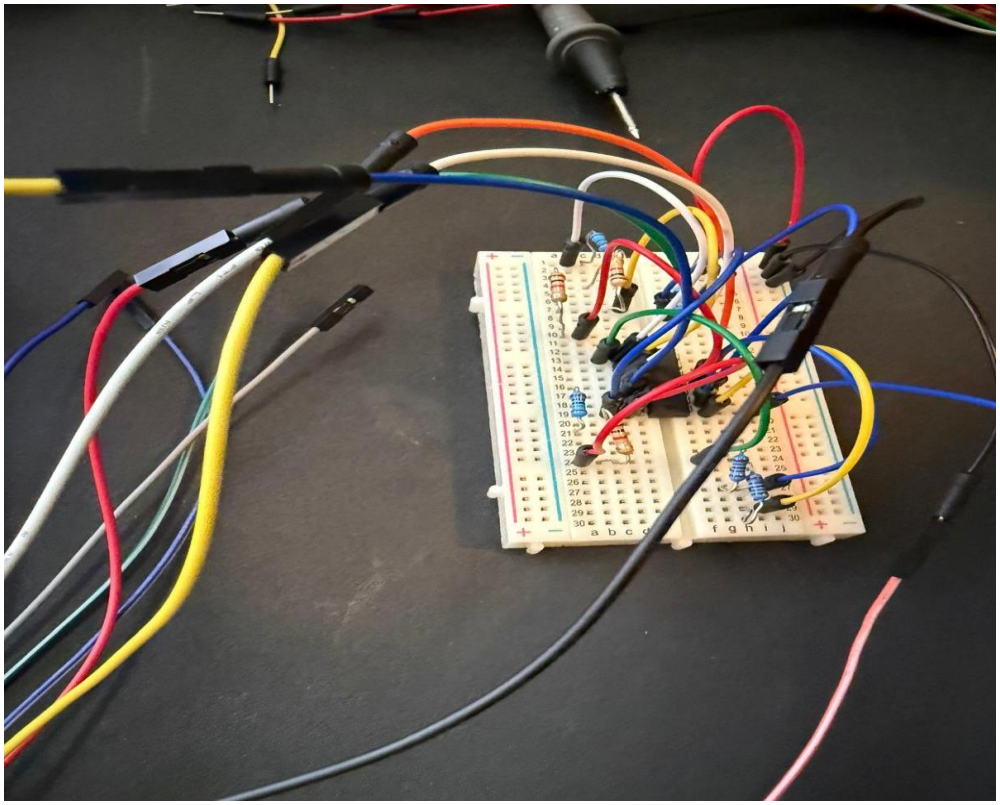

Figure 46. Stage 1 on breadboard implementation (upper view)

Figure 47. Output voltage when differential input voltage amplitude is maximum

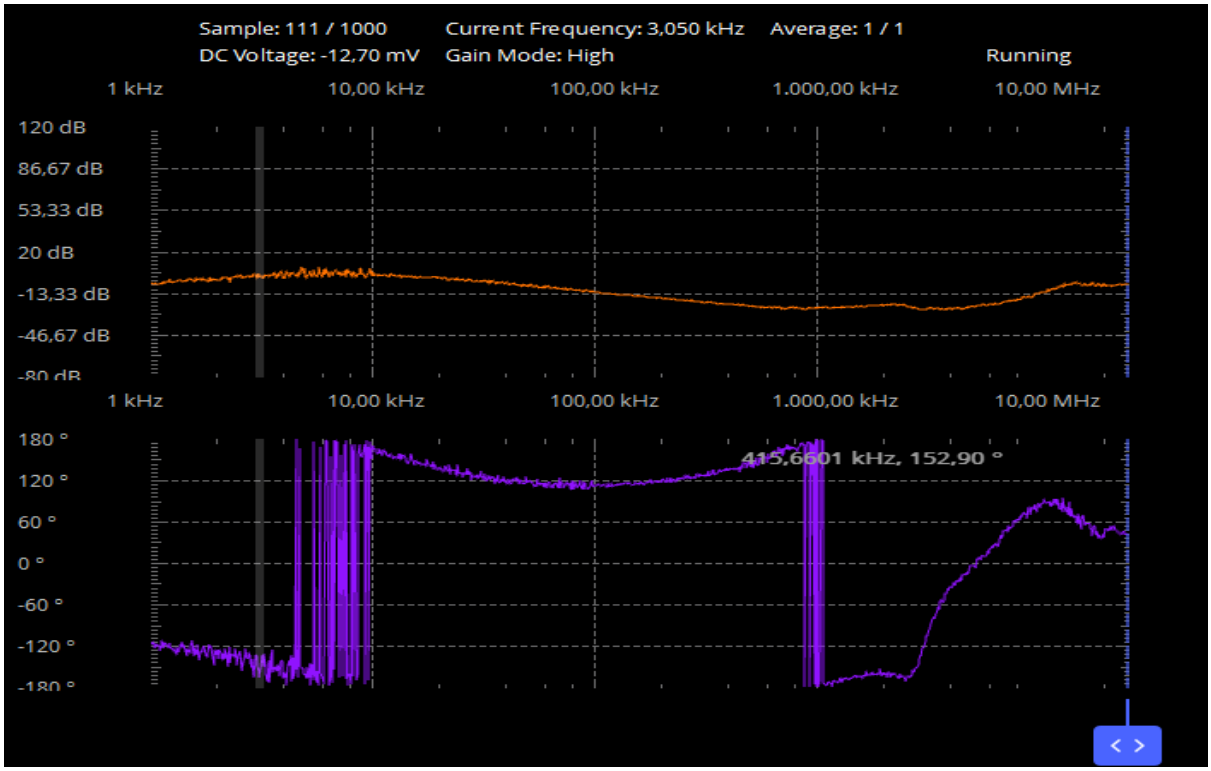

Figure 48. Bode characteristics

Because in LTSpice the circuit was simulated with a high-precision and high-speed ADA4627 operational amplifier and a different power supply, and for the practical implementation of the same circuit a much slower and much less precise operational amplifier OP482 was used, it does not behave as we would expect, as can be seen in Figure 48. However, a gain of approximately 15dB at a frequency of 7kHz is obtained, no parameters obtained in LTSpice can be extracted from here because we did not obtain a good frequency characteristic.

Figure 49. Output voltage when differential input voltage amplitude is maximum and step-type

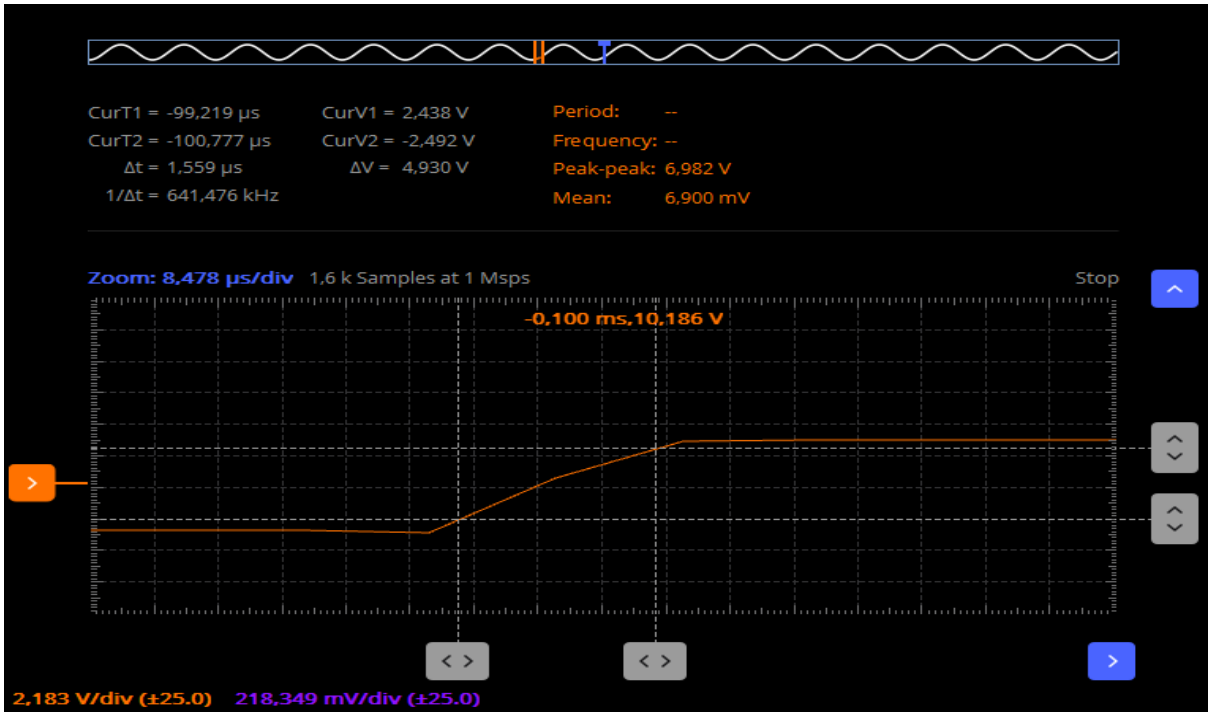

Figure 50. Output voltage slew rate

If we apply a step-type signal with very small rise and fall times (ns) to the input, figure 50 shows a slew rate of approximately 3V/µs..

SR = **ΔV/ΔT** = 5V/1.6µs = 3.125V/µs

2. Stage 2

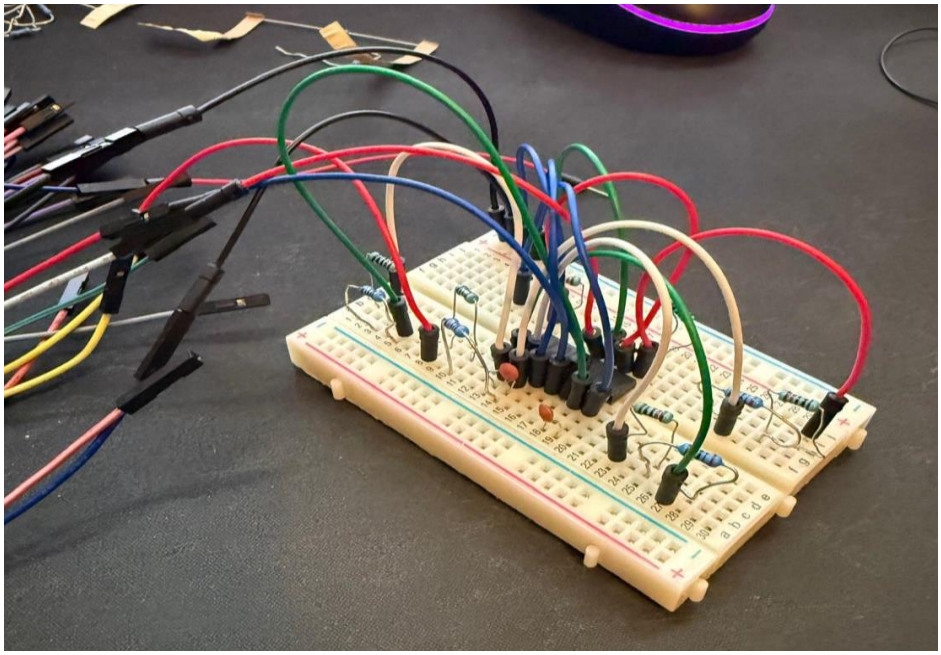

Figure 51. Stage 2 on breadboard implementation (side view)

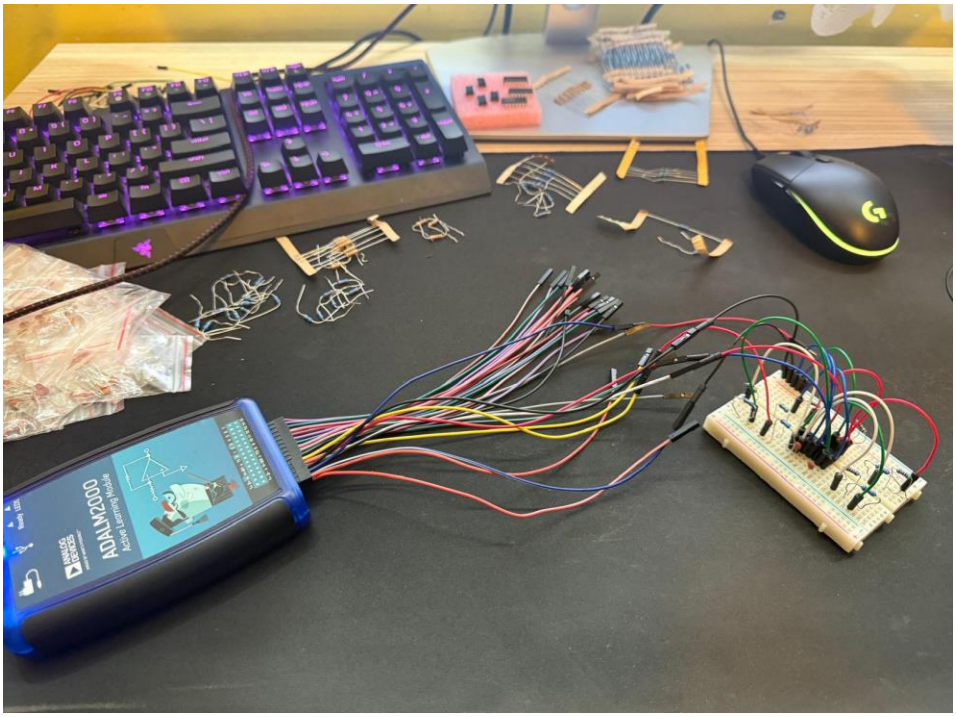

Figure 52. Stage 2 on breadboard implementation (distant view)

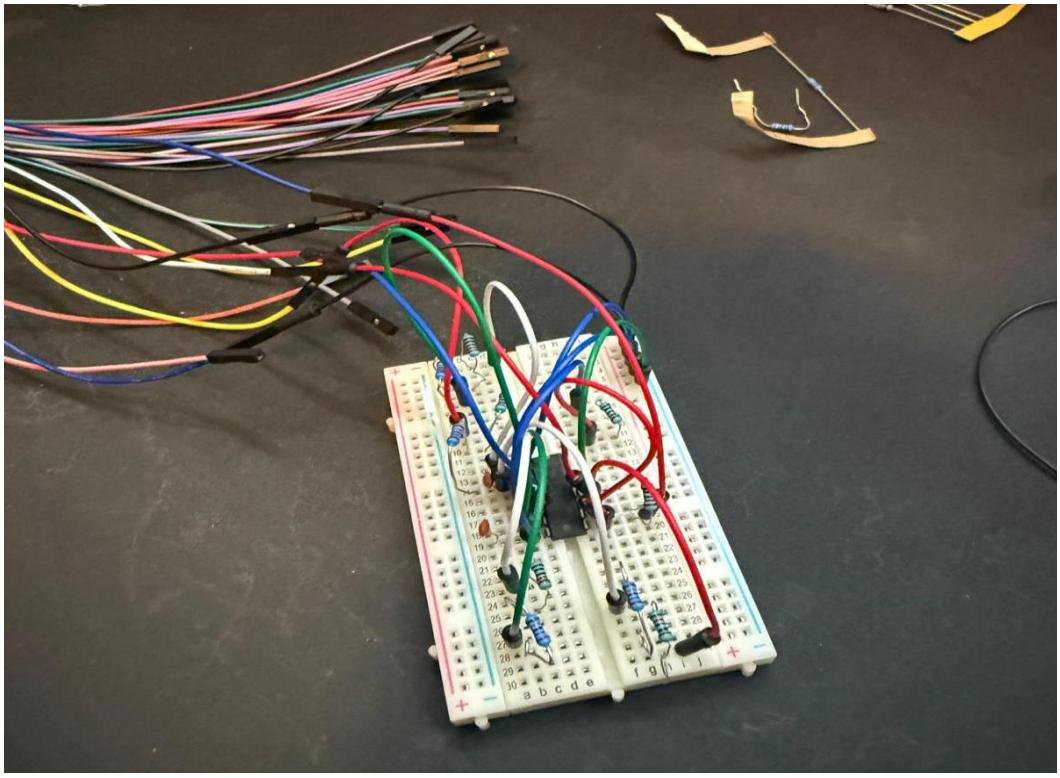

Figure 53. Stage 2 on breadboard implementation (upper view)

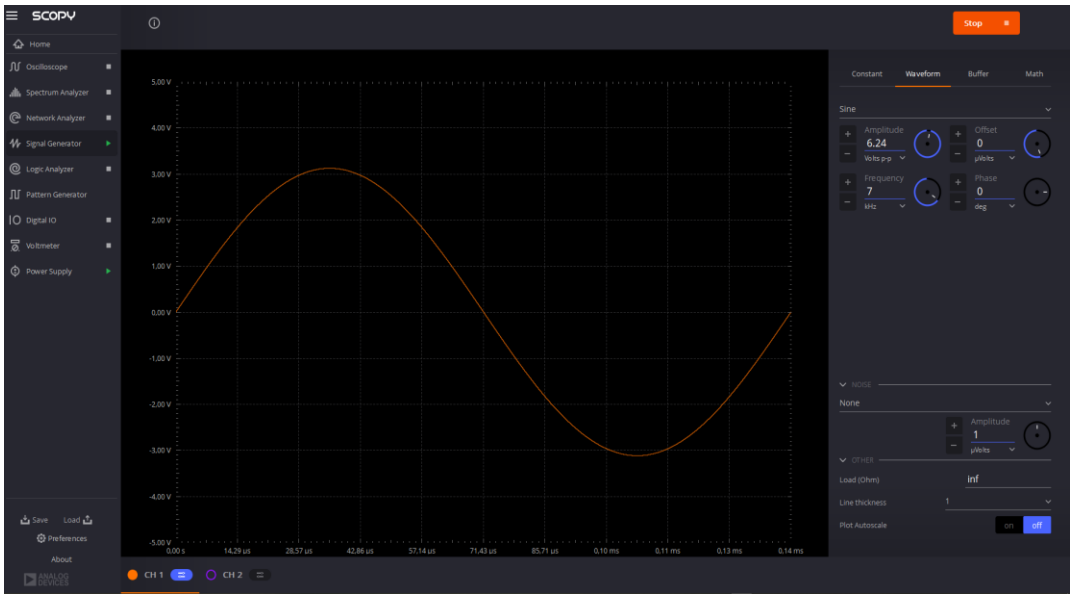

Figure 54. Software-generated input voltage signal

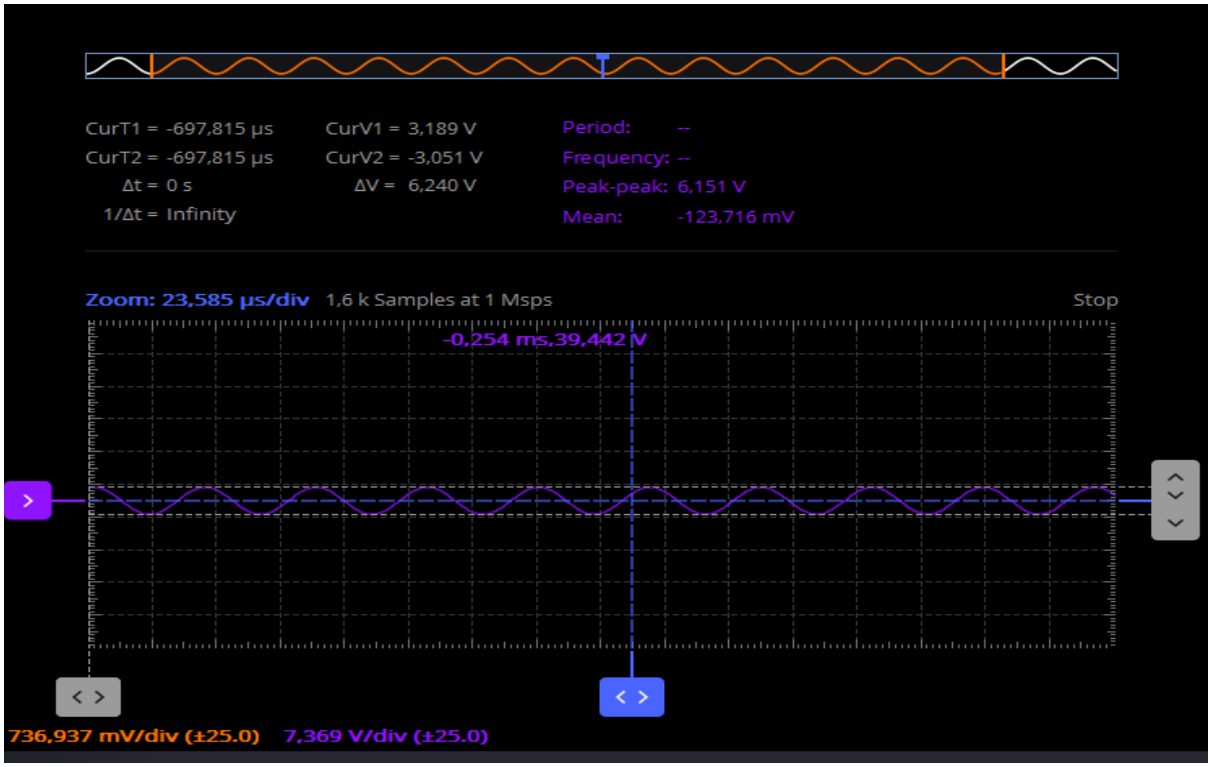

Figure 55. Output voltage signal on oscilloscope

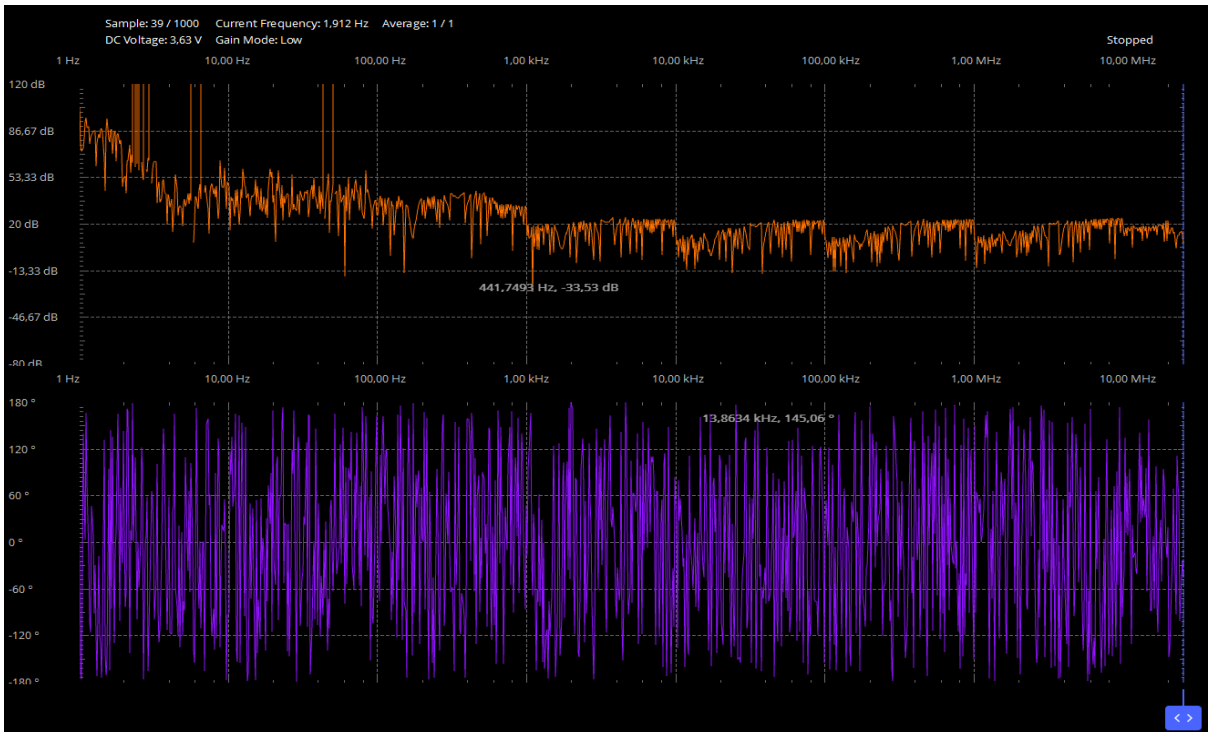

Figure 56. Bode characteristics

We cannot practically extract the parameters obtained in LTSpice because we did not obtain a good filter characteristic in Scopy. However, the output signal has approximately the same amplitude as the input signal, so we can deduce the unity gain in the passband.

3. Stage 3

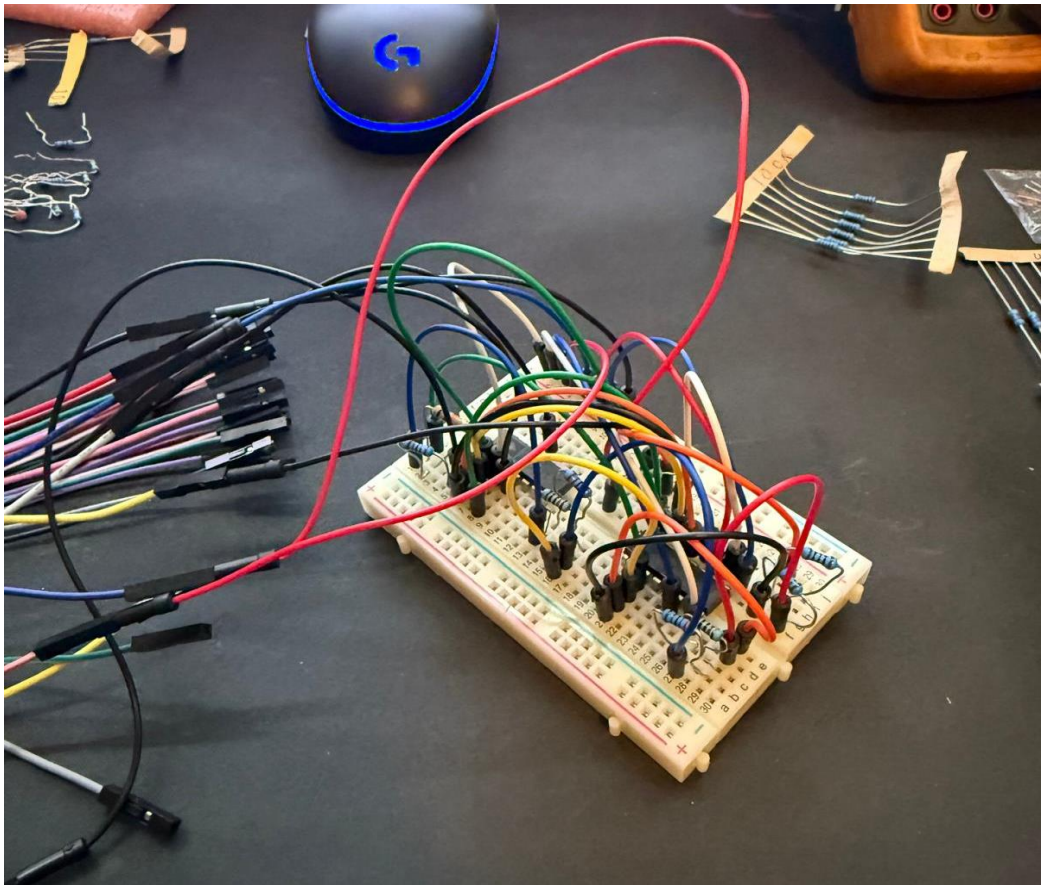

Figure 57. Stage 3 on breadboard implementation (side view)

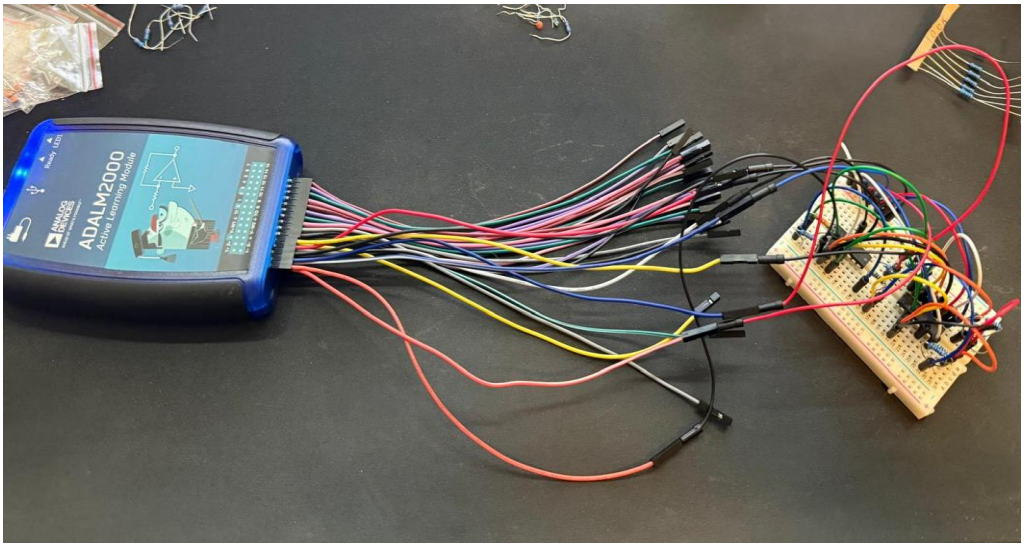

Figure 58. Stage 3 on breadboard implementation (distant view)

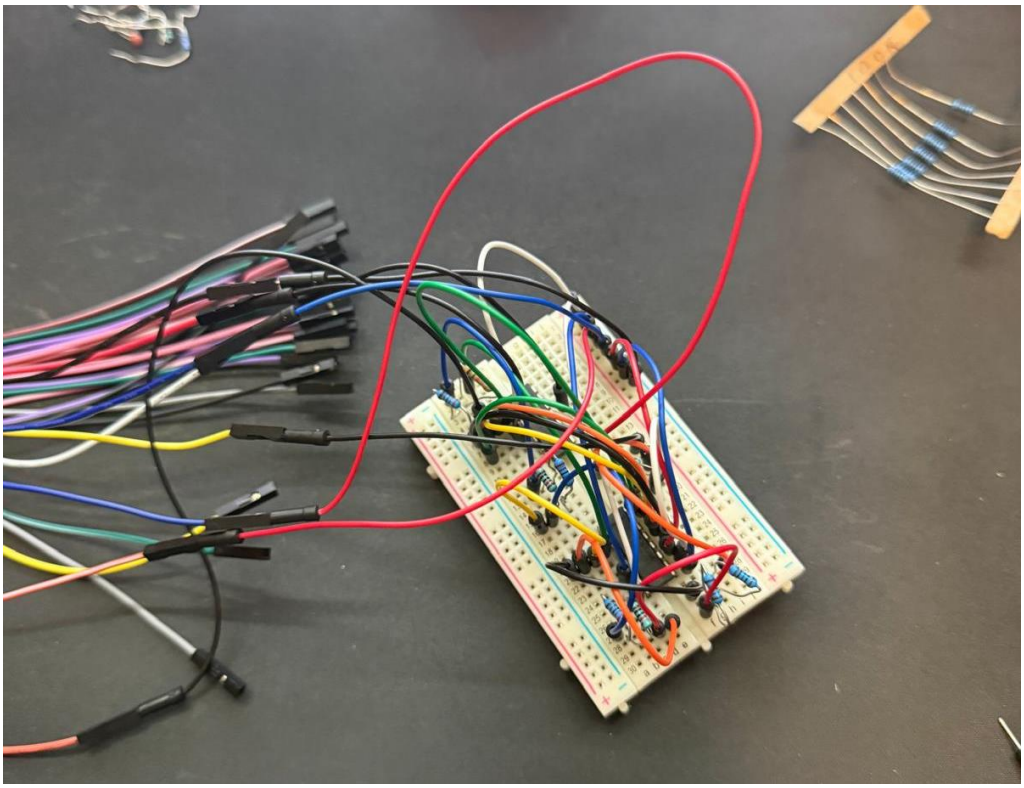

Figure 59. Stage 3 on breadboard implementation (upper view)

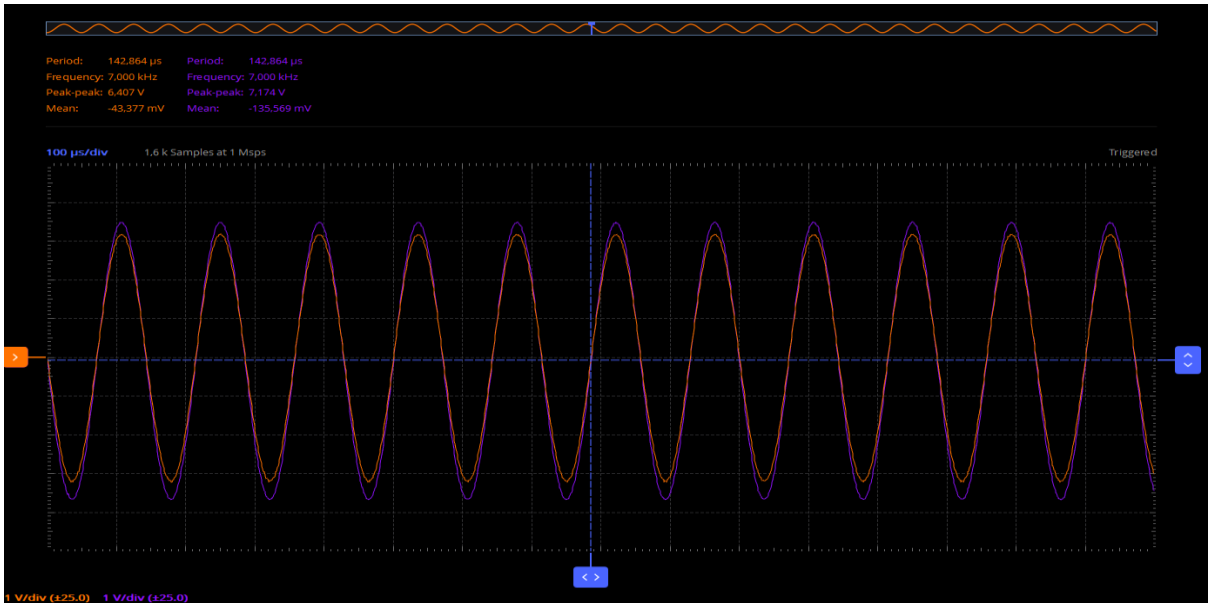

Figura 60. V_in (orange) (A_vin = 3.12V) and V_out (purple) for 1dB PGA gain

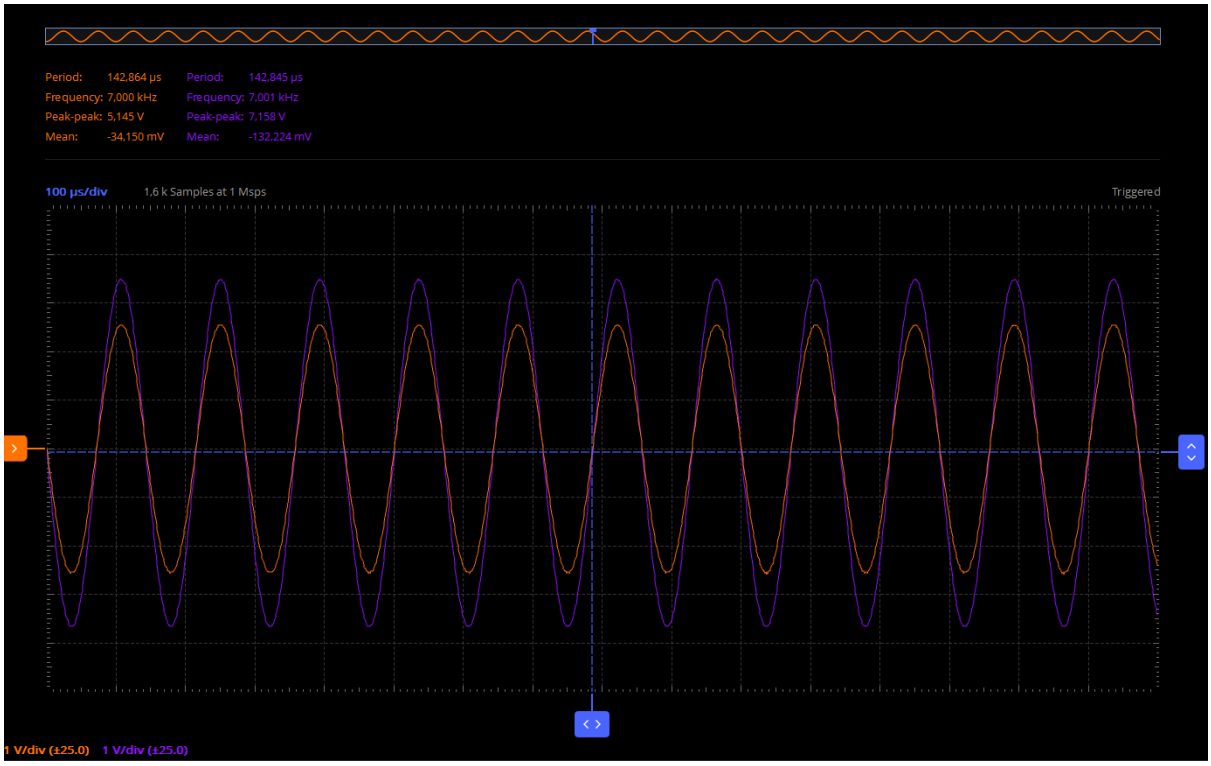

Figura 61. V_in (orange) (A_vin = 2.5V) and V_out (purple) for 3dB PGA gain

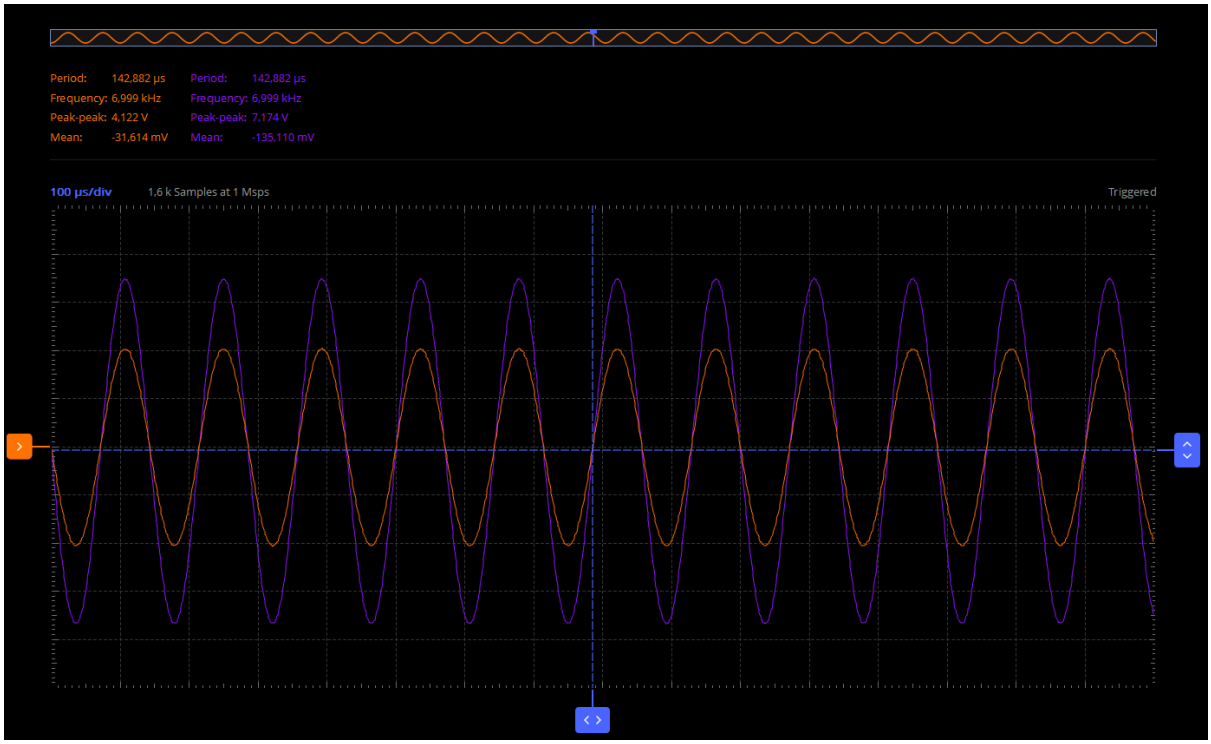

Figure 62. Figura 61. V_in (orange) (A_vin = 2V) and V_out (purple) for 5dB PGA gain

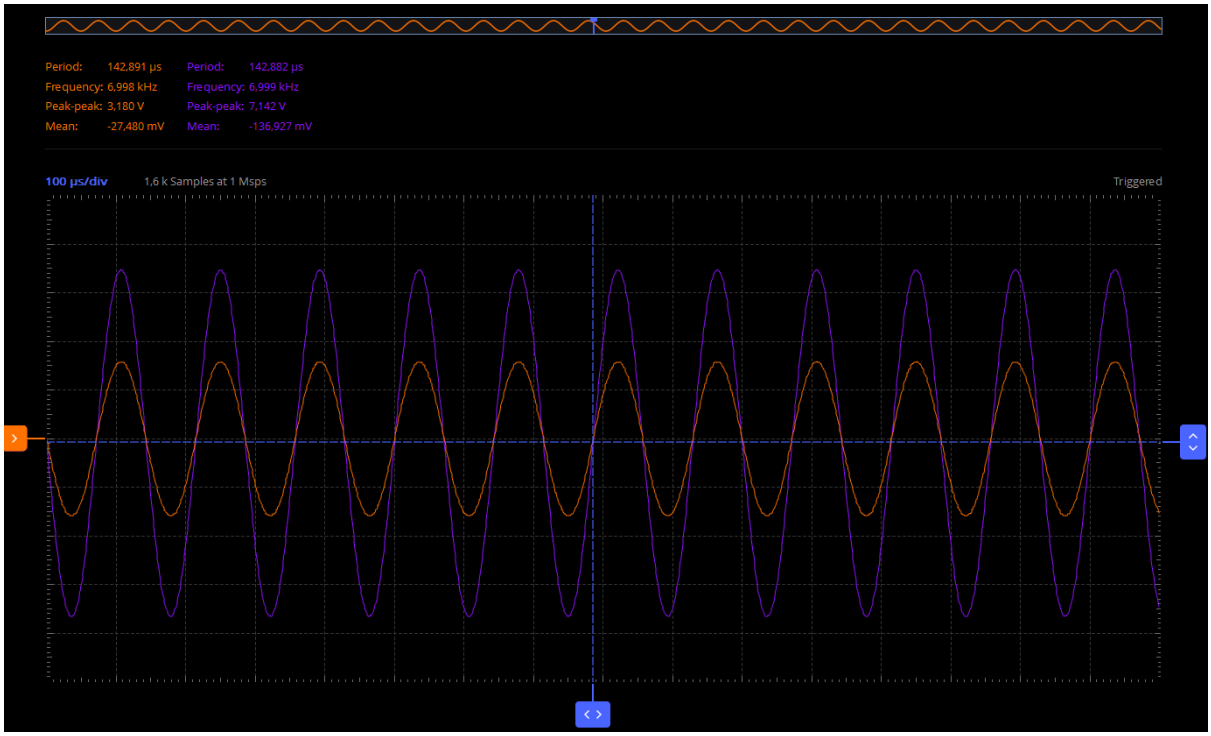

Figure 63. Figura 61. V_in (orange) (A_vin = 1.56V) and V_out (purple) for 7dB PGA gain

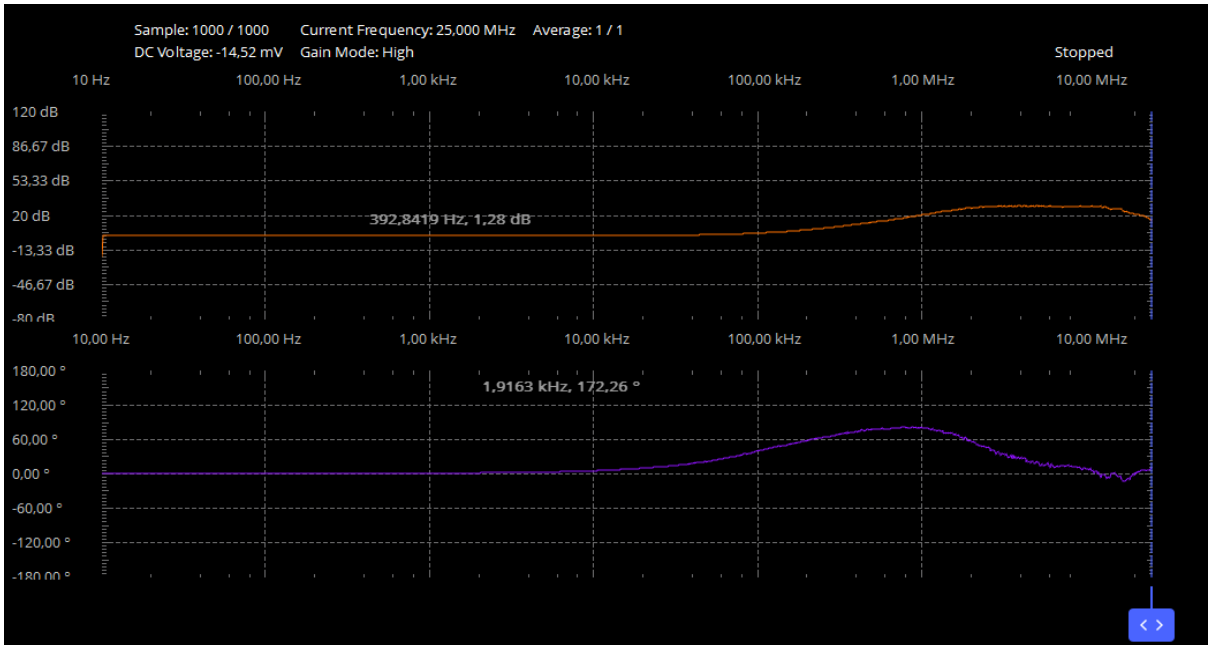

Figure 64. Bode characteristics

For all 4 gains 1dB, 3dB, 5dB and 7dB the Bode characteristics look identical only that they have the specific gain on the magnitude characteristic. Thus, practically no parameters obtained in LTSpice can be extracted because we also did not obtain a good frequency characteristic in this case.

4. Stage 4

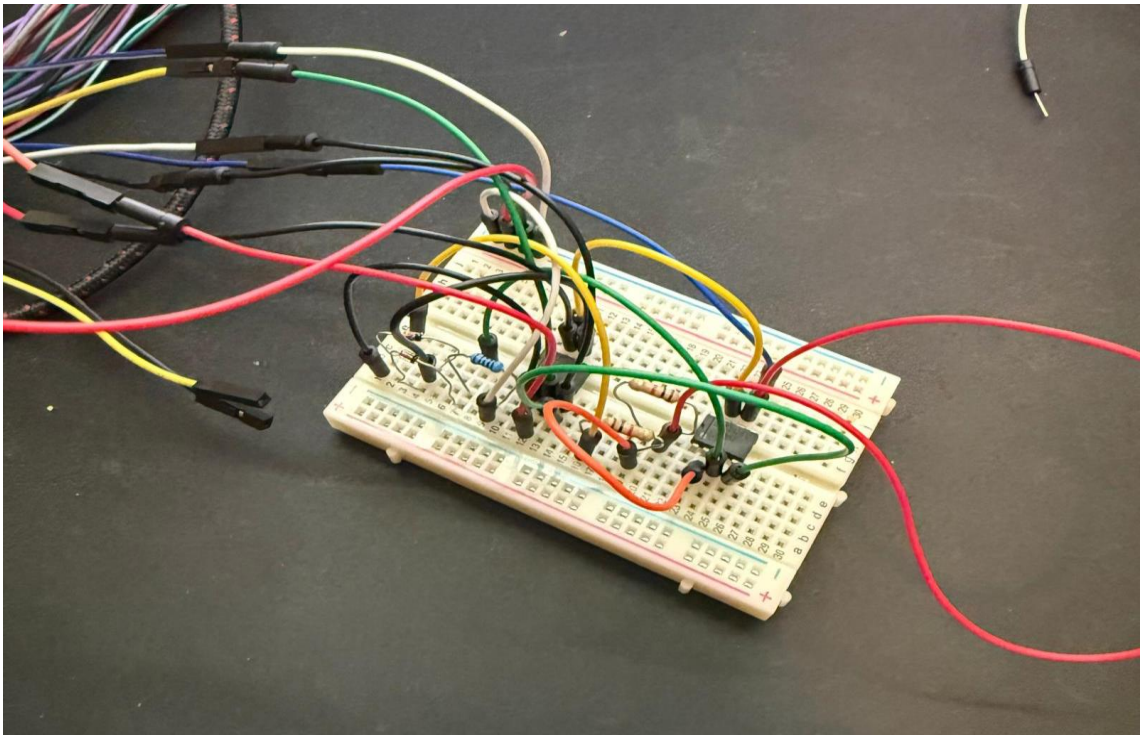

Figure 65. Stage 4 on breadboard implementation (upper view)

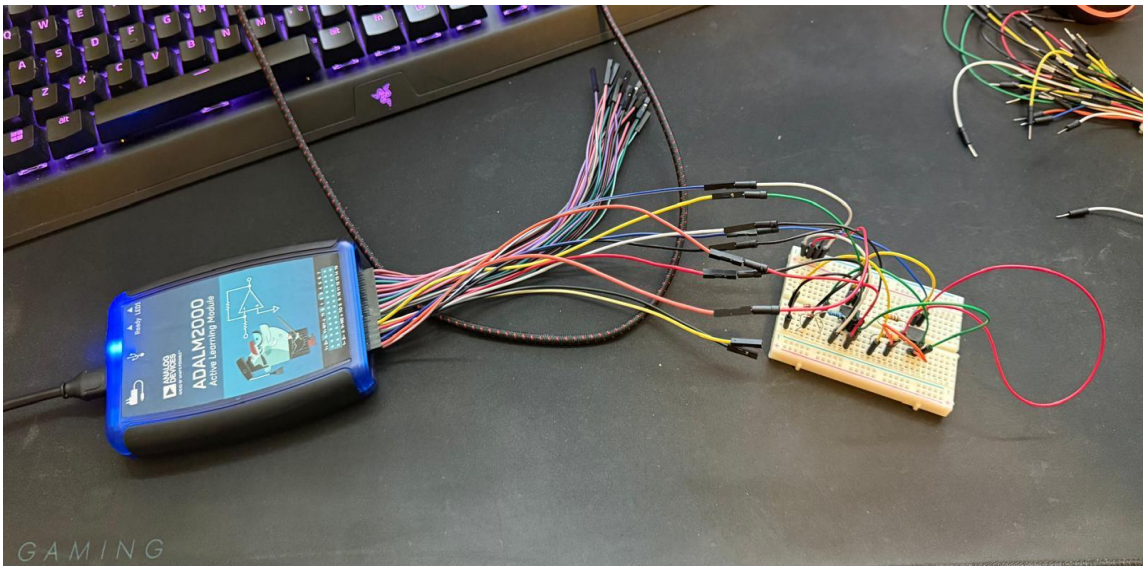

Figure 66. Stage 4 on breadboard implementation (distant view)

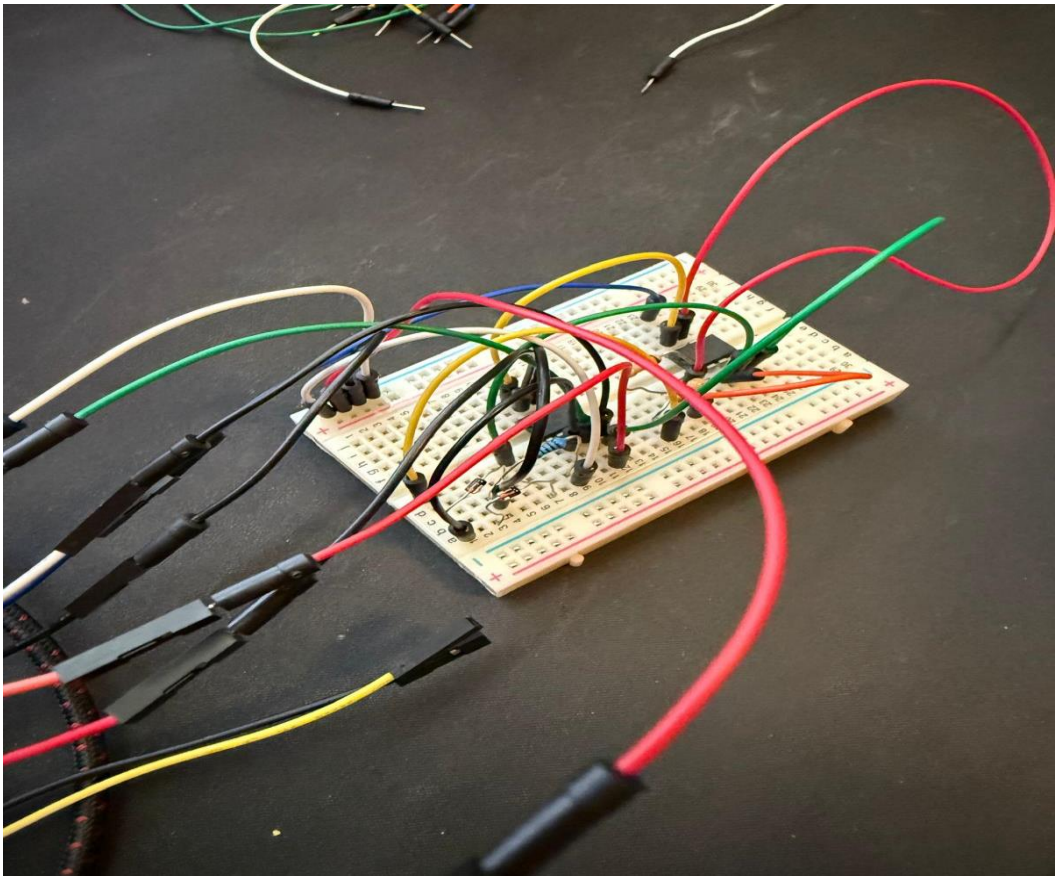

Figure 67. Stage 4 on breadboard implementation (side view)

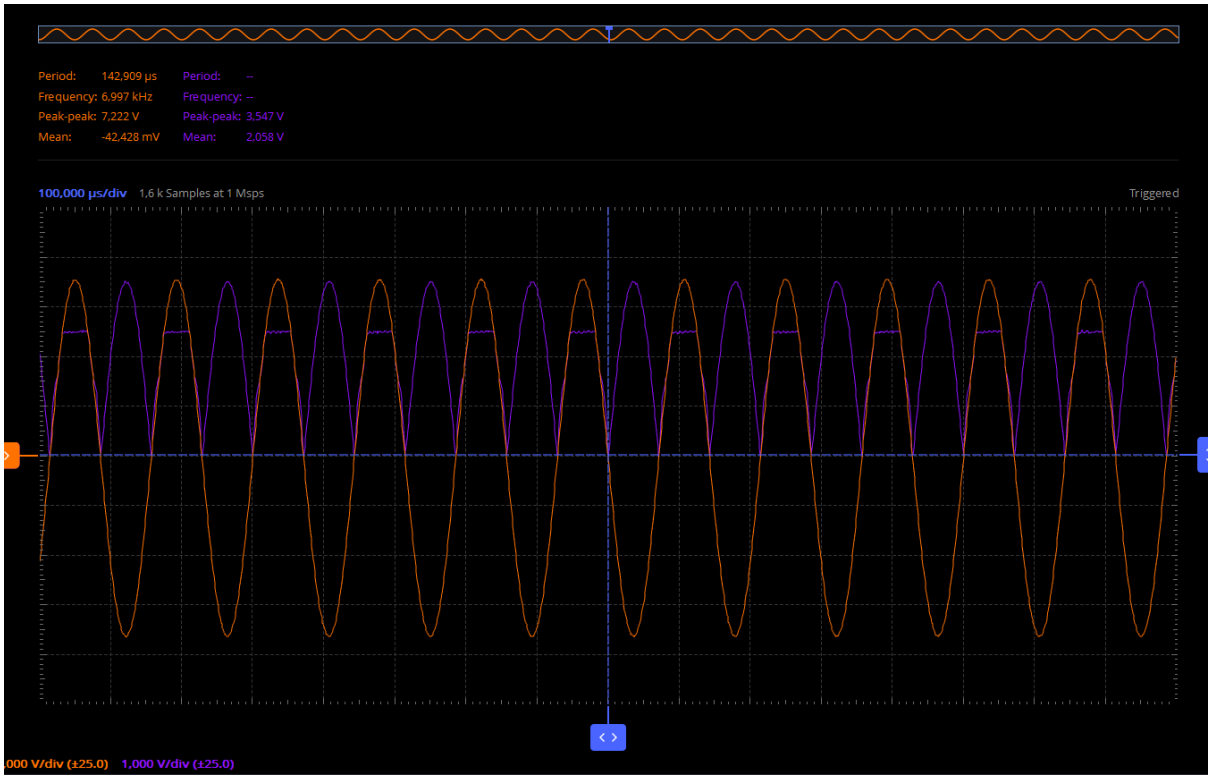

Figure 68. Full-wave rectifier (input signal – orange, output signal – purple)

According to the simulations in LTSpice, the rectification on the positive alternation is limited to 3.5V amplitude for the input signal and +/-5V supply due to the output voltage swing of OA, which also happens on the practical circuit as seen in figure 68.